

练习题-定积分定义及含定积分的极限

1. 计算下列极限

$$(1) \lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt{1 + \cos \frac{\pi}{n}} + \sqrt{1 + \cos \frac{2\pi}{n}} + \cdots + \sqrt{1 + \cos \frac{n\pi}{n}} \right);$$

$$(2) \lim_{n \rightarrow \infty} \left(\frac{1}{n\sqrt{n+1}} + \frac{\sqrt{2}}{n\sqrt{n+\frac{1}{2}}} + \cdots + \frac{\sqrt{n}}{n\sqrt{n+\frac{1}{n}}} \right);$$

$$(3) \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{i}{(n + \sqrt{i})^2} \ln \left(1 + \frac{i}{n} \right);$$

$$(4) \lim_{n \rightarrow \infty} \frac{1}{n + \sqrt[3]{n} + 100} \sum_{k=1}^n \arctan \frac{2k-n}{n};$$

$$(5) \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n (2k-1) \sin \frac{2k-1}{2n};$$

$$(6) \lim_{n \rightarrow \infty} \left(\frac{n^2+1}{n^3+1} + \frac{n^2+2}{n^3+2^3} + \cdots + \frac{n^2+n}{n^3+n^3} \right);$$

$$(7) \lim_{n \rightarrow \infty} \frac{\sqrt[n]{(n+1)(n+2)\cdots(n+n)}}{n}.$$

2. 设螺旋 $r = \theta, 0 \leq \theta \leq 2\pi$ 与极轴所围区域的面积为 A , 则 $A = ()$

$$(A) \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{4\pi^3 i^2}{n^3}; \quad (B) \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{4\pi^3 i^2}{n^2}; \quad (C) \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{8\pi^3 i^2}{n^3}; \quad (D) \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2\pi^3 i^2}{n^3}.$$

$$3. \text{ 计算极限 } \lim_{x \rightarrow +\infty} \sqrt[3]{x} \left[\ln \left(1 + \frac{\ln 2}{\sqrt[3]{x}} \right) + \int_x^{x+1} \frac{\sin t}{\sqrt{t + \cos t}} dt \right].$$

$$4. \text{ 计算极限 } \lim_{x \rightarrow 0} \int_{\ln(1+x)}^x \frac{(1-2t)^{\frac{1}{t}}}{t^2} dt.$$

5. 设 $S(x) = \int_0^x |\cos t| dt$, (1) 当 n 为正整数, 且 $n\pi \leq x \leq (n+1)\pi$ 时, 证明:

$$2n \leq S(x) \leq 2(n+1); \quad (2) \text{ 求 } \lim_{x \rightarrow +\infty} \frac{S(x)}{x}.$$

6. (1) 证明: 当 $0 \leq t \leq 1$ 时, $2t \leq 1 + \sin \frac{\pi t}{2} \leq 2$; (2) 求 $\lim_{n \rightarrow \infty} \left[\int_0^1 \left(1 + \sin \frac{\pi t}{2} \right)^n dt \right]^{\frac{1}{n}}$.

7. (1) 证明: 当 $0 \leq x \leq 1$ 时, $0 \leq \ln(1+x) \leq x$; (2) 求 $\lim_{n \rightarrow \infty} \int_0^1 |\ln x| [\ln(1+x)]^n dx$.

8. 证明 $\lim_{n \rightarrow \infty} \int_0^{\frac{\pi}{2}} \sin^n x dx = 0$ (n 为正整数).

9. 设 $I_n = \int_0^1 \frac{x^n}{1+x} dx, n=1,2,\dots$, (1) 求极限 $\lim_{n \rightarrow \infty} nI_n$; (2) 求极限 $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{(-1)^k}{k}$.